

Problem Statement: Optimizing RAG Document Retrieval for Agronomic Advice

Access to agricultural information remains a challenge for smallholder farmers. Although extension manuals, seminars, webinars, and the internet provide access to agricultural information, physical manuals are often limited, expensive, outdated, difficult to obtain, and time-consuming to search. General search engines also return broad or irrelevant results because they are not optimized for agricultural domains. These limitations can hinder access to reliable information and negatively affect agricultural decision-making and productivity.

This project aims to develop an offline retrieval engine that identifies and ranks the most relevant documents from a knowledge base of agricultural extension materials in response to farmers' questions. The system will cover key agronomic topics, including crop production, diseases, pests, and soil management, across twelve common African crops. It will return the top five documents relevant to each query, enabling faster and more effective access to agricultural information.

REFERENCES

Clarke, J. I. (2026, July 23). Fruits and vegetables. In Encyclopedia Britannica.
<https://www.britannica.com/place/Africa/Fruits-and-vegetables>

Empowering farmers with artificial intelligence: a retrieval-augmented generation based large language model advisory framework (2026) Journal of Agricultural Engineering, 57(2).
doi:10.4081/jae.2026.1908.

Sawant, S., Nair, R., & Hariharan, S. (2026). Empowering farmers with artificial intelligence: A retrieval-augmented generation based large language model advisory framework. Journal of Agricultural Engineering, 57, Article 1908. <https://doi.org/10.4081/jae.2026.1908>